Applications of exponential and logarithmic functions

Compound interest, half-lives, earthquakes and sound — the places where these two functions are not optional.

LESSON 67–68 | 2 × 40 MIN | ALGEBRA 10, §3.9 | P2 · 2.1, 2.5

LESSON CLOCK

16' | 22' | 22' | 20' | 18' | 10'

Growth, and solving for the time	16 min
Decay and half-life	22 min
Logarithmic scales	22 min
Linearising	20 min
Practice	18 min
Homework	10 min

LEARNING OBJECTIVES

- Model compound growth and decay, and solve for the time.
- Use a logarithm to bring an unknown out of an index.
- Interpret a logarithmic scale (pH, decibels, Richter).
- Linearise an exponential model by taking logarithms.

TEXTBOOK

National	pp. 285–296
Cambridge	Pure Mathematics 2 & 3, pp. 26–32, 47–52
Workbook	P2 Exercise 2H

01 Explanation

Growth, and solving for the time

$$A = P(1 + r)^n$$

P the starting amount, r the rate per period as a decimal, n the number of periods. Finding A is arithmetic; finding n needs a logarithm:

$$n = \frac{\lg(A/P)}{\lg(1 + r)}$$

Example. 5 million so‘m at 14% a year. When does it double?

$$1.14^n = 2 \Rightarrow n = \frac{\lg 2}{\lg 1.14} = \frac{0.3010}{0.0569} \approx 5.29 \text{ years}$$

WHY THE LOGARITHM IS NEEDED

The unknown is in the index. No amount of rearranging brings it down — only taking a logarithm of both sides does. That is the practical reason this whole chapter exists.

Decay and half-life

$N = N_0 \cdot (½)^\wedge(t/h)$ where h is the half-life

Solving for t again needs a logarithm:

$$t = h \cdot \frac{\lg(N/N_0)}{\lg 0.5}$$

SUBSTANCE	HALF-LIFE	USE
carbon-14	5730 years	dating organic remains
iodine-131	8 days	medical imaging
caffeine in the body	about 5 hours	why coffee keeps you awake

Logarithmic scales

When a quantity ranges over many powers of ten, its logarithm is the readable number:

SCALE	DEFINITION	ONE UNIT MEANS
Richter	$M = lg(\frac{A}{A_0})$	10 times the amplitude
decibel	$L = 10\ lg(\frac{I}{I_0})$	10 dB is 10 times the intensity
pH	$pH = -lg[H^+]$	10 times the concentration

An earthquake of magnitude 7 is not slightly worse than one of 6 — it is 10 times the amplitude and about 32 times the energy.

Linearising

Given data believed to follow $y = k \cdot a^x$, take logarithms of both sides:

$$lg\ y = lg\ k + x \cdot lg\ a$$

Plotting $lg\ y$ against x gives a **straight line** of gradient $lg\ a$ and intercept $lg\ k$. So a ruler on the transformed plot finds both parameters — and a straight plot is itself the evidence that the model is exponential.

For a power law $y = k \cdot x^n$, plot $lg\ y$ against $lg\ x$ instead: the gradient is then n.

02 Worked examples

EXAMPLE 1 **5 million so‘m is invested at 14% a year. How long until it doubles?**

1. $1.14^n = 2$
2. $n\ lg\ 1.14 = lg\ 2$
3. $n = \dfrac{0.3010}{0.0569}$
4. ≈ 5.3 years.

ANSWER

≈ 5.3 years

EXAMPLE 2 **A sample has half-life *8* days. What percentage remains after *20* days?**

1. $(½)^{(20/8)} = (½)^{2.5}$

2. $= \dfrac{1}{4\sqrt{2}} \approx 0.1768$

3. $\approx 17.7\%$

ANSWER

$\approx 17.7\%$

EXAMPLE 3 **One earthquake has magnitude *6.4*, another *7.9*. Compare their amplitudes.**

1. $M = lg\,\dfrac{A}{A_0}$

2. $7.9 - 6.4 = 1.5$

3. $10^{1.5} \approx 31.6$

ANSWER

About *32* times greater

03 Interactive model

Ask for the doubling time of any growth rate mentioned in the news.

Bringing the unknown down

To solve $1.14^n = 2$ you should:

- divide by 1.14
- take logarithms
- square it
- guess

5 million at 14%: doubling time is about:

- 3 y
- 5.3 y
- 7 y
- 14 y

Half-life 8 days: after 24 days there remains:

- $\frac{1}{3}$
- $\frac{1}{8}$
- $\frac{1}{16}$
- $\frac{1}{24}$

One Richter unit is a factor of:

- 2
- 10
- 100
- 32

To find a in $y = k\,a^x$ from data, plot:

- y vs x
- $lg\,y$ vs x
- $lg\,y$ vs $lg\,x$
- y vs $lg\,x$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Compound interest	Murakkab foiz	Сложные проценты
Growth factor	O‘sish koeffitsienti	Коэффициент роста
Half-life	Yarim yemirilish davri	Период полураспада
Doubling time	Ikkilanish vaqti	Время удвоения
Logarithmic scale	Logarifmik shkala	Логарифмическая шкала
Richter scale	Rixter shkalasi	Шкала Рихтера
Decibel	Detsibel	Децибел
Linearisation	Chiziqlashtirish	Линеаризация
Model parameter	Model parametri	Параметр модели

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Compound interest is modelled by:

- $P + rn$
- $P(1 + r)^n$
- $P \cdot rn$
- $P + r^n$

To bring an unknown out of an index:

- divide
- take a logarithm
- square
- differentiate

Two half-lives leave:

- $\frac{1}{2}$
- $\frac{1}{4}$
- $\frac{1}{3}$
- nothing

For a power law plot:

- $lg\ y$ vs x
- $lg\ y$ vs $lg\ x$
- y vs x
- y vs $lg\ x$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 1000 at 10% for 3 years

   ANSWER 1331

2. 2000 at 5% for 2 years

   ANSWER 2205

3. Half-life 4 days; fraction after 8 days

   ANSWER $\frac{1}{4}$

4. Half-life 4 days; fraction after 12 days

   ANSWER $\frac{1}{8}$

5. One Richter unit is a factor of

   ANSWER 10

6. 10 dB is a factor of

   ANSWER 10 in intensity

7. pH 4 versus pH 6: how many times more acidic?

   ANSWER 100

Medium · the standard question

7 PROBLEMS

1. 5 million at 14%: doubling time

   ANSWER ≈ 5.3 years

2. $20\ 000$ at 8%: when does it reach $50\ 000$?

   ANSWER ≈ 11.9 years

3. Half-life 8 days: percentage after 20 days

   ANSWER $\approx 17.7\%$

4. Magnitudes 6.4 and 7.9: amplitude ratio

   ANSWER ≈ 32

5. A population trebles in 12 years. Annual rate?

   ANSWER $\approx 9.6\%$

6. A car falls 18% a year. When is it worth half?

   ANSWER ≈ 3.5 years

7. Sound at 85 dB versus 65 dB: intensity ratio

   ANSWER 100

Hard · stretch and reason

7 PROBLEMS

1. Carbon-14 half-life 5730 y; 30% remains. Age?

   ANSWER ≈ *9950* years

2. P at r compounded monthly: the annual growth factor

   ANSWER $(1 + \frac{r}{12})^{12}$

3. *10%* compounded monthly versus annually on *1000* for 1 year

   ANSWER *1104.71* versus *1100*

4. Data fits $y = k\,a^x$ with $lg\ y$ line of gradient 0.3 and intercept 1. Find k, a

   ANSWER $k = 10, a \approx 2$

5. Data fits $y = k\,x^n$ with $lg\ y$ vs $lg\ x$ gradient 1.5, intercept 0.6. Find k, n

   ANSWER $n = 1.5, k \approx 3.98$

6. Show the doubling time at rate r is about $\frac{70}{100r}$

   ANSWER The "rule of 70" from $\frac{\ln 2}{r}$

7. A drug halves every 5 h. What fraction remains after 24 h?

   ANSWER ≈ *3.6%*

07 Homework

Homework — 6 tasks

Every answer needs its units and a sentence in the words of the question.

1. *8* million so‘m at *12%* a year. Find the value after 6 years and the doubling time.
2. A sample has half-life *6* hours. Find the percentage remaining after *20* hours.
3. Two earthquakes measure *5.8* and *7.3*. Compare their amplitudes.
4. A population of *12 000* grows *3%* a year. When does it pass *20 000*?

5. Data believed to follow $y = k a^x$ gives a $lg\ y$ line of gradient 0.48 and intercept 0.7. Find k and a.
6. Explain in three sentences why a logarithmic scale is used for earthquakes.